An Average Is Not a Guarantee: Making Assumptions Visible

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Abstract

A bound on an aggregate is easy to overstate when its assumptions disappear between a proof and an abstract. This educational note restates the elementary range bound for a convex combination, proves it directly, and gives a signed- weight counterexample. Nonnegative weights that sum to one keep a weighted average within the range of its inputs. The result does not establish that every input is accurate, that the average is unbiased, or that a predictive model generalizes. The purpose is to demonstrate assumption-aware theoretical writing, not to present a new theorem or a conference submission.

1 The Writing Problem

The sentence "aggregation guarantees reliable predictions" mixes a mathematical property with an empirical promise. Even when an aggregate satisfies a range bound, nothing in that bound establishes that the inputs are close to an unknown target. The proof and the headline need to answer the same question.

We use a convex combination, a standard object in convex analysis [Boyd and Vandenberghe, 2004], to separate the two statements. The argument below is written independently for this example. It is elementary and carries no novelty claim.

2 Setting and Proposition

Let $y_1, \ldots, y_n$ be real numbers, with finite $n \geq 1$. Let $w_i \geq 0$ for every i, and suppose $\sum_{i=1}^{n} w_i = 1$. Define the aggregate

$$\bar{y} = \sum_{i=1}^{n} w_i y_i.$$

Proposition. Under these assumptions,

$$\min_i y_i \;\leq\; \bar{y} \;\leq\; \max_i y_i.$$

The statement is deterministic. It does not assume a training distribution, sampling independence, or a probabilistic model. Conversely, it does not provide any conclusion about those quantities.

3 Proof and Assumption Use

Write $m = \min_i y_i$ and $M = \max_i y_i$. For every index, $m \leq y_i \leq M$. Since w_i is nonnegative, multiplication preserves both inequalities: $w_i m \leq w_i y_i \leq w_i M$. Summing over indices gives

$$m \sum_i w_i \;\leq\; \sum_i w_i y_i \;\leq\; M \sum_i w_i.$$

Normalization, $\sum_i w_i = 1$, gives the claimed interval. This identifies where each assumption enters: nonnegativity preserves order, while normalization returns the bounds to the original scale.

For unnormalized nonnegative weights with positive total W, the same argument applies to w_i/W, and hence to $(\sum_i w_i y_i)/W$. If $W = 0$, that expression is undefined. Dropping the denominator is not an equivalent estimator and does not inherit the same range bound.

4 Counterexamples to Broader Claims

Normalization alone is insufficient. Take $(y_1, y_2) = (0, 1)$ and $(w_1, w_2) = (-1, 2)$. The weights sum to one, but the aggregate is 2, outside the input interval $[0, 1]$. Negative weights invalidate the order argument in the proof. A manuscript cannot omit nonnegativity while retaining the proposition unchanged.

Nor does the range bound imply accuracy. Suppose two predictors both output 100 when the true value is 0. Every convex combination of those outputs remains 100. The proposition holds exactly, yet the squared error is 10000. There is no contradiction: containment and accuracy are different properties.

The example also gives no guarantee of unbiasedness. Bias requires a target and an expectation under a specified distribution, neither of which appears in the proposition. A paper claiming such a guarantee would need additional assumptions and a separate argument.

5 Verification and Limits

The repository tests finite rational examples using exact arithmetic and checks the counterexamples numerically. Those tests are useful for catching transcription mistakes; they do not prove the proposition for all real inputs. The proof above supplies that argument. This distinction is the theoretical analogue of separating a measured result from a universal empirical claim.

This note contains no new aggregation method, empirical benchmark, formal proof-assistant verification, or peer-review outcome. It was authored with AI assistance as an open writing example. Its scientific scope is the stated elementary inequality and the demonstrated limits of its interpretation.

6 Conclusion

The defensible headline is that nonnegative normalized aggregation preserves the input range. Reliability, accuracy, and generalization are not consequences of that statement alone. A theoretical manuscript should keep the assumptions that make its claim true visible in the abstract, proposition, and interpretation.

References

Stephen Boyd and Lieven Vandenberghe. *Convex Optimization.* Cambridge University Press, 2004. URL https://web.stanford.edu/~boyd/cvxbook/.